Name: ______________________

Worksheet

Investigating transformations with Desmos

Task A: Investigating translation with Desmos

Draw any polygon and translate it to the right using a vector.

1) Click on your object and drag it around the workspace.
   What happens to the image? Why does it do this?

2) Click on the end of your vector and drag to make the vector longer.
   What happens to the image? Why does it do this?

3) Click on the middle of your vector and drag it around the workspace.
   What happens to the image? Why does it do this?

Task B: Investigating rotation with Desmos

Draw any polygon, rotate it using a slider and set the slider to 270°.

1) Drag the centre of rotation towards the image.
   What happens to the image? Why does it do this?

2) Is it possible for the centre of rotation to be inside the image?
   If so, how? If not, why not?

3) Move the spot in the slider left and right. For what angles does the image perfectly overlay the object? Why does it does this for these angles?

Task C: Investigating reflection with Desmos

Draw a polygon where every edge is a different length (e.g. a scalene triangle).
Draw a vertical line and reflect the polygon over the line.

1) Drag the line of symmetry towards the image.
   What happens to the image? Why does it do this?

2) Is it possible for the line of symmetry to be inside the image?
   If so, how? If not, why not?

3) Is it possible for the image to perfectly overlay the object?
   If so, how? If not, why not?

Task D: Investigating enlargement with Desmos

Draw any polygon and enlarge it with a slider scale factor.

1) Drag the slider left and right.
   a) When is the image larger than the object?

   b) When is the image the same size as the object?

   c) When is the image smaller than the object?

   d) What happens to the image when the scale factor is 0? Why does this happen?

2) Select the line tool.
   Click on a vertex on the object and its corresponding vertex on the image to draw a line that traverses both vertices. Repeat for all pairs of corresponding vertices.
   Where do the lines intersect each other?